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Notes and comments

Concentrations of carbon dioxide in the cabin of a small passenger car

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ABSTRACT

This communication examines carbon dioxide levels in a small passenger car in city driving. It looks at situations with only a driver and when an adult passenger is on board; in both cases with and without air ventilation on.

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1. The Experiment

This paper is a follow-up to an investigation concerning air quality in an estate car (Gładyszewska-Fiedoruk, forthcoming) with the experimental set-up being modified so that the experiment better reflects typical driving practice. Thus the tests were conducted in a small passenger car with a cabin capacity of 2.5 m^3. Such small vehicles usually serve as a means of transport for one or two people. The experiment consisted of looking at the effects of having one or two persons in the car and of each four ventilation system settings. The measurements took place during a 90-min drive in Bialystok; 90 min being about the average time spent daily in a car by a driver and passenger.

Before the experiment commenced, effectiveness of the car ventilation system was tested for all the available air inflow intensities. The system was set to provide air through four vents, each one with the surface of 0.0028 m^2, the vents being located thus; vent 1 – by the window, left of the driver's seat; vents 2 and 3 – in the middle of the dashboard: vent 2 to the right of the driver's seat, vent 3 to the left of the front passenger's seat; and vent 4 – by the window, right of the front passenger's seat. When the cabin air circulation was switched off, air flow measurements were carried out for all available air inflows over a 90 min.

In the case when the cabin air circulation was on, depending on the number of people travelling, the measurements lasted 45 for one person and 30 min for two. After these periods, the concentration of carbon dioxide inside the cabin was about. 4500 ppm. Subsequently, the air circulation was switched off and the cabin was being aired by the car's ventilation system, and the experiment was continued until 90 min was reached.

Initially, it was assumed that when driving a car a person exhales 35 g/h of CO_2, the same as when doing light sedentary work (Figs. 1 and 3 – calculations35). This assumption was corrected so that the analytical calculations would match the experiment results (Figs. 1 and 3 – calculations66). Further, all the series included the measurement of concentrations of carbon dioxide flowing into the cabin through the vents. On this basis, a balance of carbon dioxide concentration inside the cabin was worked out. Finally, temperature and humidity of both cabin air, and the air flowing inside through the vents were measured, although we concentrate on CO_2.

2. Results

Air parameters were recorded using a testo435 measuring instrument, five being taken every time, and their mean values calculated. The experiment was carried out under conditions involving initial air parameters seen in Table 1. Figs. 1 and 2

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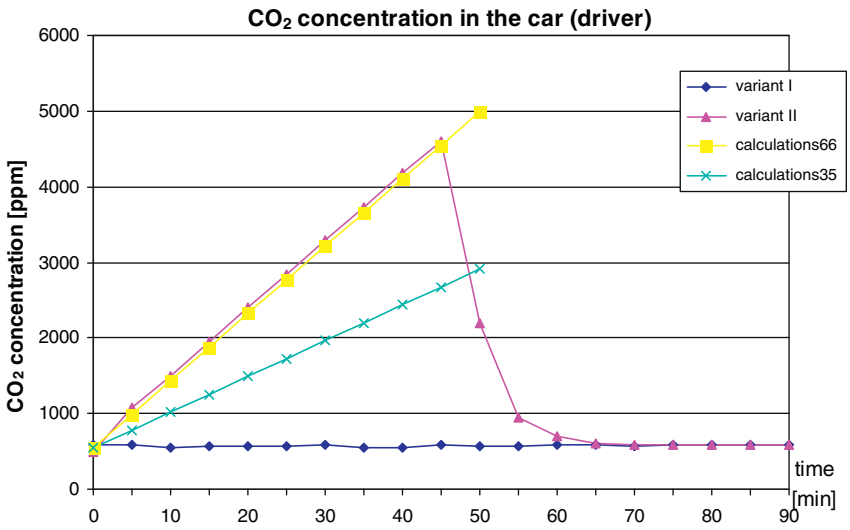

Fig. 1. CO_2 concentration values in the cabin (driver only) – cabin air circulation on.

Table 1
Car ventilation system efficiency tests.

Measuring point	Velocity of the incoming airflow (m/s)				Expenditure (air inflow intensity)		Air change rate (h^{-1})
	1	2	3	4	(m^3/s)	(m^3/h)	
0	0.28	0.41	0.25	0.27	0.0034	12.32	4.93
1	2.88	2.56	3.15	2.90	0.0325	116.95	46.78
2	3.97	3.41	4.36	3.99	0.0445	160.11	64.04
3	6.06	5.36	6.46	6.25	0.0682	245.61	98.25
4	8.26	7.42	8.88	8.70	0.0940	338.55	135.42

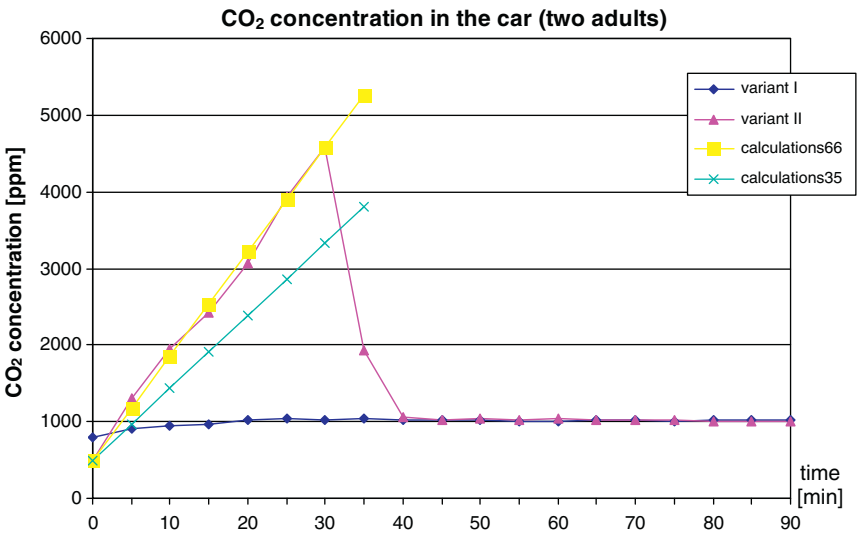

Fig. 2. CO_2 concentration values in the cabin (two adults) – cabin air circulation on.

show CO_2 concentrations in the car during the 90-min drive. These values were recorded with the ventilators in the car operating in all available modes (i.e. with all air inflow intensities). The figures, however, illustrate the results only for mode 1 – the smallest velocity of inflowing air, that is 117 m^3/h. Fig. 1 concerns the first part of the experiment, when only the driver is in the car, whereas Fig. 2 pertains to the second part, with the driver and one passenger in the car. Fig. 3 shows

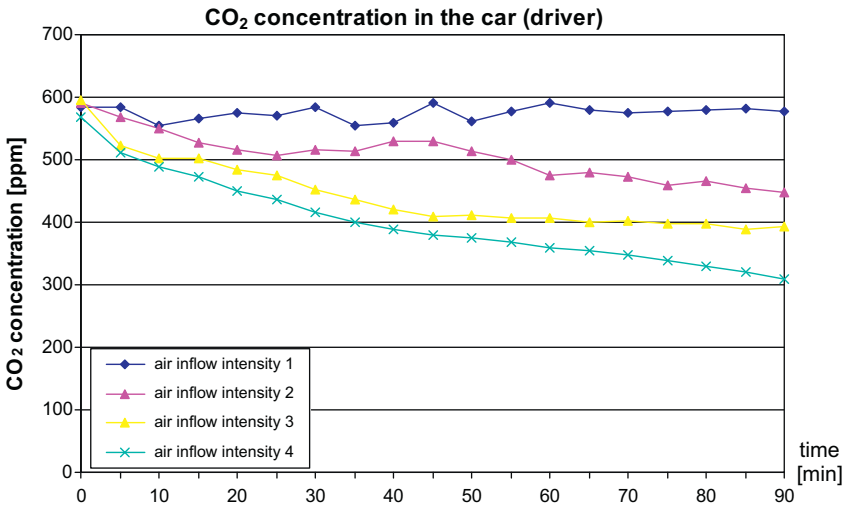

Fig. 3. CO_2 concentration values in the cabin (driver only) – cabin air circulation off.

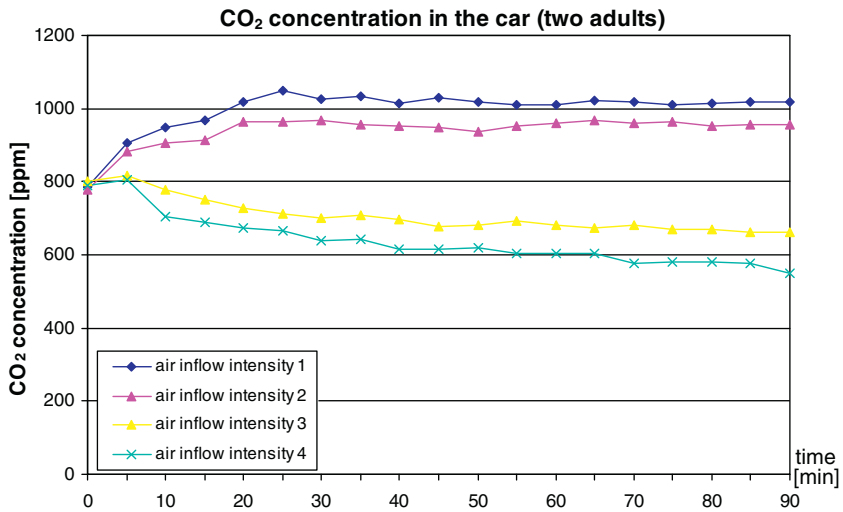

Fig. 4. CO_2 concentration values in the cabin (two adults) – cabin air circulation off.

the distribution of CO_2 concentrations in the car with inactive interior air circulation and for all available air inflow intensities when only the driver is in the car. Similarly, Fig. 4 presents the same data for the situation with two people in the car.

Table 2 contains data relating to the distribution of CO_2 concentration in the car cabin with inactive interior air circulation and for all available air inflow intensities when only the driver is in the car. Table 2 shows CO_2 concentration balance for both parts of the experiments with operating cabin air circulation, and for air inflow intensity set to one.

3. Discussion

The intensity of inflowing air is usually dictated by the driver and by the passenger, and most often the ventilation control is set to one or two. During the experiment, air was flowing into the cabin through four vents, but even though the ventilation system was centrally regulated, each of the vents had different outputs (Table 1). The air change rate was high with nearly five changes when the mechanical ventilation was inactive, and up to 135 changes when the ventilation was fully working. Also, air flowed in through the leaks in the cabin.

Fig. 1 gives results with ventilation control set to one (117 m^3/h). Variant I represent the measurements taken with inactive cabin air circulation; in this case, the values of CO_2 concentration oscillate around the initial value of 580 ppm during experiment. Analogously, Variant II represents measurement taken with the cabin air circulation operat-

Table 2
Distribution of CO_2 inside the cabin with air circulation on.

Time (min)	Air inflow intensity			
	1 – 117 m^3/h	**2** –160 m^3/h	**3** – 245 m^3/h	**4** – 338 m^3/h
0	596	562	582	577
5	1073	1145	1045	1101
10	1491	1562	1662	1552
15	1945	1901	1951	1940
20	2411	2367	2395	2402
25	2833	2789	2801	2803
30	3287	3190	3206	3209
35	3737	3582	3622	3694
40	4177	3999	4035	4207
45	4598	4388	4457	4511
50	2201	1975	1834	1612
55	941	911	874	767
60	706	688	632	576
65	609	576	508	522
70	548	517	517	503
75	508	509	499	487
80	500	511	507	474
85	501	500	500	474
90	508	506	498	467

ing. During the first 45 min the CO_2 concentration level reached 4598 ppm, but after the interior air circulation had been switched off, with other settings remaining the same, the concentration sharply dropped to 2201 ppm in the next 5 min and later steadily fell to 561 ppm in the end of the experiment. The calculations66 line results can from calculations for the assumed amount of exhaled carbon dioxide equal to 66 g/h; an amount estimated to be exhaled during physical exercise and the Similarly, the calculations35 line shows calculations for assumed exhaled carbon dioxide of 35 g/h;, an amount exhaled during light sedentary work and the value used as a basis for calculating the amount of CO_2 exhaled by drivers.

Fig. 3 shows results with inactive cabin air circulation, for all the four air inflow intensities. A downward trend is can be noticed for all the lines, and with the air inflow intensity set to four the CO_2 concentration is almost the same as in outdoor air. On the other hand, the line representing air inflow intensity set to one fluctuates around the initial value. For the settings examined the quality of cabin air remained within the recommended limits by the European Branch of World Health Organization (2000), ASHRAE (1989), and the Polish standard PN-EN 13779 (2008).

The complete results regarding CO_2 concentrations for all available air inflow intensities, and with activated cabin air circulation, cannot be presented in a graphic form because the overlap. In fact, during the initial 45 min of the experiment, the values of CO_2 concentration differed only slightly regardless of the air inflow intensity (Table 2). The differences were, however, observed during the subsequent airing of the cabin because of the different outputs of ventilators. With the air inflow intensity set to four, at the end of the experiment the CO_2 concentration fell to that typical of outdoor air. Furthermore, in when the cabin air circulation was active, after 5 min of driving the quality of cabin air dropped below recommended values. Figs. 2 and 4 consider the situation when there are two people are in the car. Fig. 2 shows the results of calculations conducted for air inflow intensity of 117 m^3/h; i.e. with ventilation control set to one. Variant I, with the interior air circulation off, illustrates that at the beginning of the experiment the concentration of CO_2 rises to about 1000 ppm., and subsequently fluctuates around this. The Variant II, with the interior air circulation on, shows that in the initial 30 min of the experiment CO_2 concentration rose to 4578 ppm and corresponds well with line representing the assumed amount of exhaled CO_2. After the cabin air circulation is switched off (other settings remaining unchanged), the concentration of CO_2 sharply falls to 1935 ppm in the first 5 min and then steadily declines to1011 ppm at the end of the experiment.

Fig. 4 presents measurements conducted with the cabin air circulation on for all the available ventilation control settings. With the air inflow intensity equal to 117 m^3/h with the ventilation control set to one, and two people in the car, the air change rate is insufficient; after 20 min of driving, CO_2 concentration exceeded 1000 ppm, and later remained constant at about 1020 ppm. Setting the ventilation control to two with air inflow intensity of 160 m^3/h, reduces the concentration to the boundary allowed by air quality standards; the CO_2 concentration fluctuated around 960 ppm. The CO_2 concentration seen in Table 3 relates to when cabin air circulation is active. In the both parts, the sharpest rise of CO_2 concentration is within the 1200–2400 ppm interval. During the airing of the car, whenever the CO_2 concentration is within this range, a rapid rise is observed.

The quality of air inside the cabin very quickly drops below recommended standards whenever the interior air circulation is turned on; with two people travelling, the permissible upper limit of CO_2 concentration is exceeded in less than three min, and after 20 min with cabin air circulation off. The ventilation system turns out to be inefficient when air inflow intensity is set to one or two.

Table 3
Carbon dioxide concentration in a car – balance sheet (ppm).

Time	One person			Two persons		
	Ventilator	Cabin	Difference	Ventilator	Cabin	Difference
0	447	496	49	415	483	68
5	1016	1073	57	1132	1305	173
10	1377	1491	114	1737	1942	205
15	1824	1945	121	2232	2426	194
20	2278	2411	133	2952	3073	121
25	2732	2833	101	3833	3944	111
30	3201	3287	86	4470	4578	108
35	3654	3737	83	1726	1935	209
40	4102	4177	75	969	1059	90
45	4521	4598	77	936	1020	84
50	2083	2201	118	967	1048	81
55	869	941	72	947	1025	78
60	641	706	65	962	1033	71
65	557	609	52	943	1015	72
70	533	578	45	956	1028	72
75	536	578	42	950	1020	70
80	543	580	37	944	1009	65
85	551	581	30	956	1011	55
90	556	578	22	965	1011	46

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